

Python Full Stack Development (Django)

Module : Python Programming Backend

Introduction to Python

- Python Installation & Setup
- Variables & Data Types
- Operators & Expressions
- Input & Output Functions
- Control Flow Statements (if-else, loops)
- Functions & Lambda Expressions
- Modules & Packages
- Lists, Tuples, Sets & Dictionaries
- String Manipulation & Regular Expressions
- File Handling (CSV, JSON, Text Files)
- Object-Oriented Programming (OOP)
- Exception Handling
- Python Libraries (NumPy, Pandas, Matplotlib, Requests)
- Database Connectivity (MySQL, SQLite)
- Web Development (Flask, Django)
- Multi-threading & Concurrency
- Data Science & Machine Learning Basics
- Web Scraping (BeautifulSoup, Scrapy)
- Automation & Scripting
- Unit Testing (unittest, pytest)
- Working with APIs & RESTful Services
- FastAPI Framework
- Deployment & Cloud Integration (Docker, AWS, GCP, Azure)

Module : Frontend Technologies

HTML

- **Introduction to HTML**
- **HTML Document Structure** (doctype, head, body, meta tags)
- **Text Formatting & Semantic Tags** (p, h1-h6, strong, em, etc.)
- **Lists & Tables** (ul, ol, dl, table, tr, th, td)
- **Forms & Input Elements** (input, textarea, select, button, label)

- **Links & Navigation** (a, nav, href, target, anchor links)
- **Images & Multimedia** (img, audio, video, iframe)
- **HTML5 New Elements** (article, section, aside, header, footer)
- **Forms & Validation** (required, pattern, minlength, maxlength)
- **CSS Integration** (inline, internal, external CSS)
- **HTML APIs** (Geolocation, Drag & Drop, Web Storage)
- **Responsive Design Basics** (meta viewport, media queries)
- **SEO Best Practices** (meta tags, alt attributes, structured data)
- **HTML Accessibility** (ARIA attributes, semantic elements)
- **Basics of JavaScript with HTML** (script tag, event handling)
- **Client Server Architecture:**

1) CSS

- CSS Syntax & Selectors (class, id, universal, attribute, pseudo-classes)
- Box Model (margin, border, padding, content)
- Flexbox (display: flex, justify-content, align-items, flex-wrap)
- Grid Layout (grid-template-rows, grid-template-columns, gap)
- Positioning (static, relative, absolute, fixed, sticky)
- Responsive Design (media queries, viewport units, rem/em)
- Typography (font-family, font-size, line-height, text-align)
- CSS Transitions & Animations (transition, @keyframes, animation)
- CSS Variables (custom properties, var())
- Backgrounds & Borders (background-image, gradients, border-radius)
- Pseudo-elements (::before, ::after)
- Z-index & Stacking Context
- CSS Filters & Effects (blur, grayscale, drop-shadow)
- CSS Frameworks Basics (Bootstrap, Tailwind CSS)
- Dark Mode & Theming (prefers-color-scheme, CSS variables)

3)JavaScripts

- JavaScript Basics (Variables, Data Types, Operators)
- Control Flow (if-else, switch, loops)
- Functions (Function Declaration, Expression, Arrow Functions, Callback Functions)
- ES6+ Features (let/const, Template Literals, Destructuring, Spread/Rest Operators)
- Objects & Arrays (Properties, Methods, Object Destructuring, Array Methods like map, filter, reduce)

- DOM Manipulation (getElementById, querySelector, Event Listeners, Modifying Elements)
- Events & Event Handling (onClick, onChange, addEventListener)
- Asynchronous JavaScript (Promises, async/await, Callbacks)
- Fetch API & AJAX (GET, POST Requests, Handling Responses)
- Closures & Scope (Global vs. Local Scope, Lexical Scope, Hoisting)
- Prototype & Inheritance (Prototype Chain, Classes, Object-Oriented JS)
- Error Handling (try-catch, throw, finally)
- Modules & Import/Export (ES6 Modules, CommonJS)
- Local Storage & Session Storage
- Event Loop & Call Stack (Execution Context, Callback Queue, Microtasks)
- Regular Expressions (RegExp) (Pattern Matching, Validation)
- Web APIs (Geolocation, Clipboard, Notifications, Speech Recognition)
- Frameworks & Libraries Basics (React.js, Vue.js, Node.js, Express.js)
- Testing & Debugging (console.log, breakpoints, Jest, Mocha)
- Best Practices & Performance Optimization (Debouncing, Throttling, Memory Leaks)

4)Advanced JavaScript

- Execution Context & Call Stack
- Scope & Closures
- Hoisting & Temporal Dead Zone (TDZ)
- Prototype & Prototypal Inheritance
- this Keyword & Binding (call, apply, bind)
- Higher-Order Functions & Functional Programming
- Event Loop, Microtasks & Macrotasks
- Promises, Async/Await & Error Handling
- Debouncing & Throttling
- JavaScript Modules (ES6 Import/Export)
- Generators & Iterators
- WeakMap & WeakSet (Memory Management)
- JavaScript Design Patterns (Singleton, Factory, Observer, etc.)
- Web Workers & Multithreading in JavaScript
- Proxy & Reflect API
- Symbol & Meta-Programming
- Service Workers & Progressive Web Apps (PWA)
- Event Delegation & Bubbling vs. Capturing
- Canvas API & WebGL for Graphics
- Testing & Debugging (Jest, Mocha, Chai, Cypress)

5)Bootstrap

- Introduction to Bootstrap (CDN, Installation, Setup)
- Bootstrap Grid System (Containers, Rows, Columns, Breakpoints)
- Typography & Text Utilities (Headings, Font Styling, Alignment)
- Bootstrap Colors & Backgrounds (Text & Background Utility Classes)
- Forms & Form Controls (Input Fields, Select, Validation, Custom Forms)
- Buttons & Button Groups (Styles, Sizes, Outline Buttons)
- Navbar & Navigation Components (Navbar, Breadcrumb, Pagination)
- Cards & Content Containers (Cards, Jumbotron, List Groups)
- Images & Media (Responsive Images, Figures, Embedding Videos)
- Modals & Popovers (Dialogs, Alerts, Tooltips)
- Bootstrap Components (Accordion, Carousel, Spinners, Progress Bars)
- Tables & Table Styling (Responsive Tables, Striped, Bordered)
- Flexbox Utilities (Alignment, Justify, Ordering)
- Bootstrap Forms Validation
- Grid System & Responsive Design
- Utility Classes (Margins, Padding, Borders, Shadows)
- Bootstrap Icons & Glyphicons
- Dark Mode & Theming with Bootstrap
- Bootstrap with JavaScript (Collapse, Dropdowns, Tabs)
- Customizing Bootstrap (SASS, Variables, Custom Themes)

Module : Database

1)SQL

- SQL Basics (Data Types, Tables, Databases)
- SELECT Statement & Data Retrieval
- WHERE Clause & Filtering Data
- ORDER BY & Sorting
- DISTINCT & Eliminating Duplicates
- Joins (INNER, LEFT, RIGHT, FULL, SELF, CROSS)
- GROUP BY & HAVING Clause
- Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- Subqueries & Nested Queries
- SQL Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT)

- Indexes & Performance Optimization
- Views (Creating & Managing Views)
- Transactions (COMMIT, ROLLBACK, SAVEPOINT)
- Stored Procedures & Functions
- Triggers & Event Handling
- Window Functions (RANK, DENSE_RANK, LEAD, LAG)
- Common Table Expressions (CTE)
- Recursive Queries
- SQL Security & User Management
- SQL Injection Prevention

2)MongoDB

- Introduction to MongoDB (NoSQL vs. SQL, Features, Use Cases)
- MongoDB Installation & Setup
- MongoDB Data Model (Documents, Collections, Databases)
- CRUD Operations (Create, Read, Update, Delete)
- MongoDB Query Language (MQL)
- Indexes in MongoDB (Single Field, Compound, Multikey, Text, Geospatial)
- Aggregation Framework (Pipeline, \$match, \$group, \$sort, \$lookup)
- MongoDB Relationships (Embedded Documents, Referenced Documents)
- Schema Design & Data Modeling
- MongoDB Transactions (ACID Compliance, Multi-Document Transactions)
- Replication & High Availability (Replica Sets)
- Sharding & Scalability (Horizontal Scaling, Shard Keys)
- MongoDB Security (Authentication, Authorization, Role-Based Access Control)
- MongoDB Performance Optimization (Indexing, Query Optimization, Caching)
- Working with MongoDB in Different Programming Languages (Node.js, Python, Java)
- MongoDB with Mongoose (for Node.js)
- MongoDB Atlas & Cloud Deployment

- Backup & Restore Strategies
- MongoDB Change Streams & Real-time Data Processing
- Monitoring & Administration (Profiler, Logs, Performance Tuning)

Module : Django Framework

- Introduction to Django
- Django Installation & Setup
- Django Project Structure
- Virtual Environments & Dependency Management
- Django Models & ORM (Object-Relational Mapping)
- Database Migrations with Django Migrations
- Django Views & URL Routing
- Django Templates & Template Inheritance
- Django Forms & Model Forms
- Django Authentication & User Management
- Django Middleware & Request/Response Lifecycle
- Class-Based Views (CBVs) vs Function-Based Views (FBVs)
- Django Static Files & Media Handling
- Django REST Framework (DRF) for APIs
- Serializers & ViewSets in DRF
- JWT & Token Authentication in DRF
- Django Signals & Event Handling
- Django Caching & Performance Optimization
- Testing in Django (Unit Tests & Integration Tests)
- Django Deployment (Gunicorn, Nginx, Docker)
- Django with AWS (S3, RDS, Elastic Beanstalk)
- Django Security Best Practices
- Django Celery for Background Tasks
- GraphQL with Django (Graphene-Django)
- Django Channels for WebSockets & Real-Time Apps